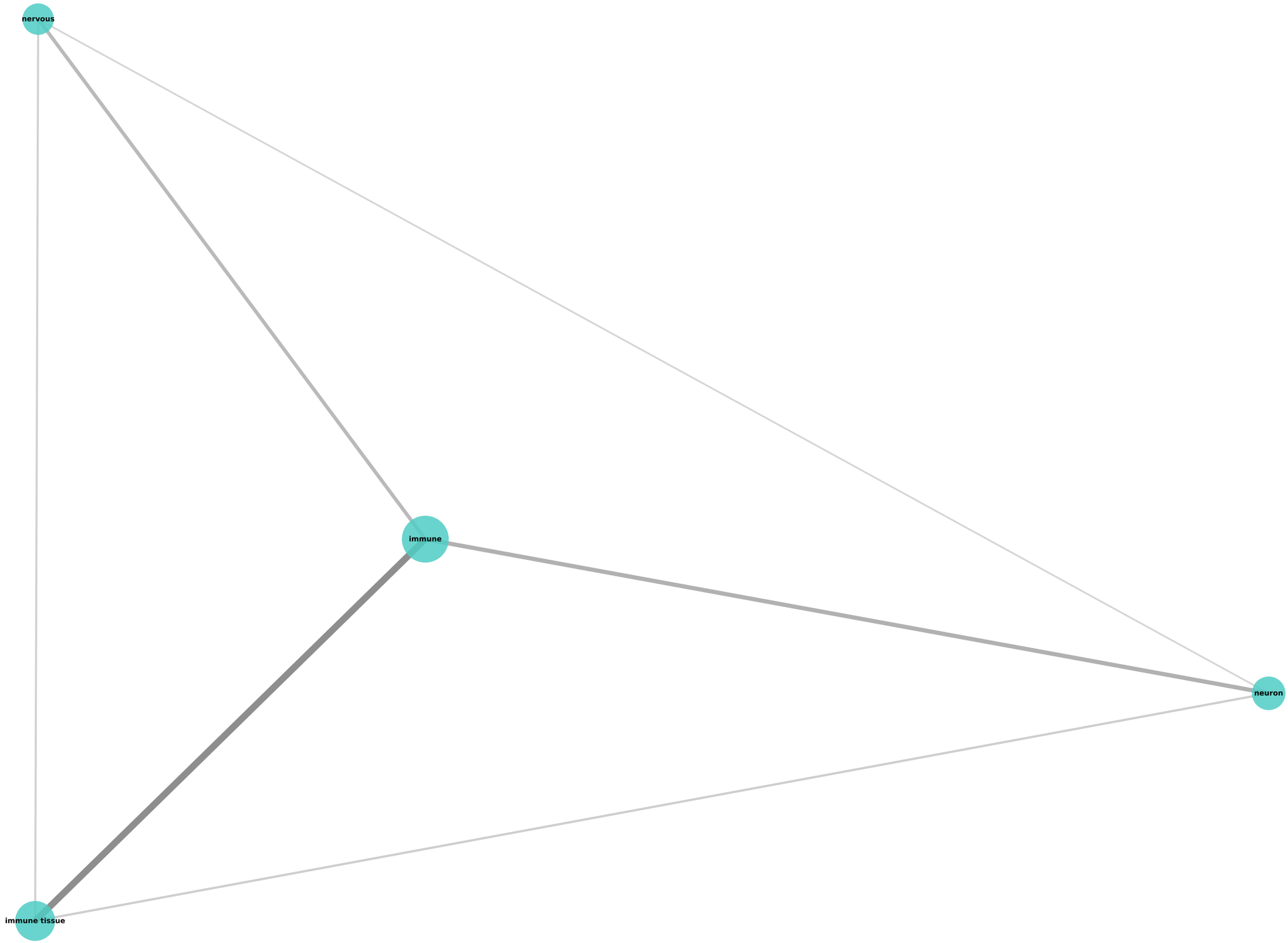Category group connectivity — pathway edges